

Worked Answer Key

1. **1:** $P = 2(8 + 3) = 22 \text{ cm}$. **2:** $P = 4 \times 6 = 24 \text{ m}$.
2. **1:** $P = 2(14 + 5) = 38 \text{ ft}$. **2:** $36 \div 4 = 9 \text{ in}$ per side.
3. **1:** $2(12 + 9) = 42 \text{ m}$. **2:** $7 + 8 + 10 = 25 \text{ in}$.
4. **1:** $11 \times 4 = 44 \text{ cm}^2$. **2:** $15 \times 6 = 90 \text{ ft}^2$.
5. **1:** $9 \times 9 = 81 \text{ yd}^2$. **2:** $12 \times 12 = 144$, so the side is **12 in**.
6. **1:** $72 \div 9 = 8 \text{ cm}$. **2:** $120 \div 8 = 15 \text{ ft}$.
7. **1:** $18 \times 7 = 126 \text{ m}^2$. **2:** One poster is $2 \times 3 = 6 \text{ ft}^2$; $4 \times 6 = 24 \text{ ft}^2$.
8. **1:** Fencing is perimeter: $2(20 + 13) = 66 \text{ yd}$. **2:** Carpet is area: $10 \times 12 = 120 \text{ ft}^2$.
9. **1:** $A = 16 \times 5 = 80 \text{ cm}^2$; $P = 2(16 + 5) = 42 \text{ cm}$. **2:** $A = 7 \times 7 = 49 \text{ m}^2$; $P = 4 \times 7 = 28 \text{ m}$.
10. **1:** $10 \times 4 + 6 \times 3 = 40 + 18 = 58 \text{ cm}^2$. **2:** $8 \times 5 + 4 \times 3 = 40 + 12 = 52 \text{ ft}^2$.
11. **1:** $5 \times 3 \times 2 = 30 \text{ cubic units}$. **2:** $7 \times 2 \times 4 = 56 \text{ cubic units}$.
12. **1:** $9 \times 4 \times 3 = 108 \text{ cm}^3$. **2:** $12 \times 5 \times 2 = 120 \text{ in}^3$.
13. **1:** Base area $8 \times 4 = 32$; $96 \div 32 = 3 \text{ m}$. **2:** Base area $6 \times 5 = 30$; $180 \div 30 = 6 \text{ ft}$.
14. **1:** $10 \times 4 \times 5 = 200 \text{ dm}^3$. **2:** $8 \times 6 \times 5 = 240 \text{ in}^3$.
15. **1:** One layer $6 \times 4 = 24$; $24 \times 3 = 72 \text{ cubes}$. **2:** $48 \times 5 = 240 \text{ cubes}$.
16. **1:** $3 \times 2 = 6 \text{ ft}^2$. **2:** $200 \text{ cm} = 2 \text{ m}$; $2 \times 3 = 6 \text{ m}^2$.
17. **1:** Garden perimeter $2(9 + 6) = 30$; $30 - 2 = 28 \text{ m}$. **2:** $4 \times 4 \times 4 = 64 \text{ cm}^3$; three length dimensions make cubic units.
18. **1:** Playground $25 \times 14 = 350$; sandbox $3 \times 4 = 12$; $350 - 12 = 338 \text{ m}^2$. **2:** $12 \times 10 \times 6 = 720 \text{ cubes}$.
19. **1:** Another pair is $6 \text{ cm} \times 8 \text{ cm}$. Its perimeter is $2(6 + 8) = 28 \text{ cm}$; the 12×4 rectangle has $P = 32 \text{ cm}$, so the new rectangle has the smaller perimeter. **2:** Original $V = 3 \times 4 \times 5 = 60 \text{ ft}^3$. Doubling length gives $6 \times 4 \times 5 = 120 \text{ ft}^3$.
20. **1:** $P = 4 \times 11 = 44 \text{ m}$; $A = 11 \times 11 = 121 \text{ m}^2$. **2:** $V = 15 \times 4 \times 3 = 180 \text{ cm}^3$; top area $= 15 \times 4 = 60 \text{ cm}^2$.